

1 Introduction

X-ray/small-angle scattering intensity $I(q)$ is a highly non-linear, 1-D fingerprint of a molecule’s 3-D shape, used throughout structural biology, materials science, and chemistry to characterize large, disordered, and dynamic molecules. The exact calculation, the Debye equation, sums every pairwise interatomic interaction at $O(N^2)$ cost, which is prohibitive for large structures. This project asks whether a GNN can learn $I(q)$ directly from atomic coordinates and element identities via a learned $O(N)$ model.

2 Illustration / Figure

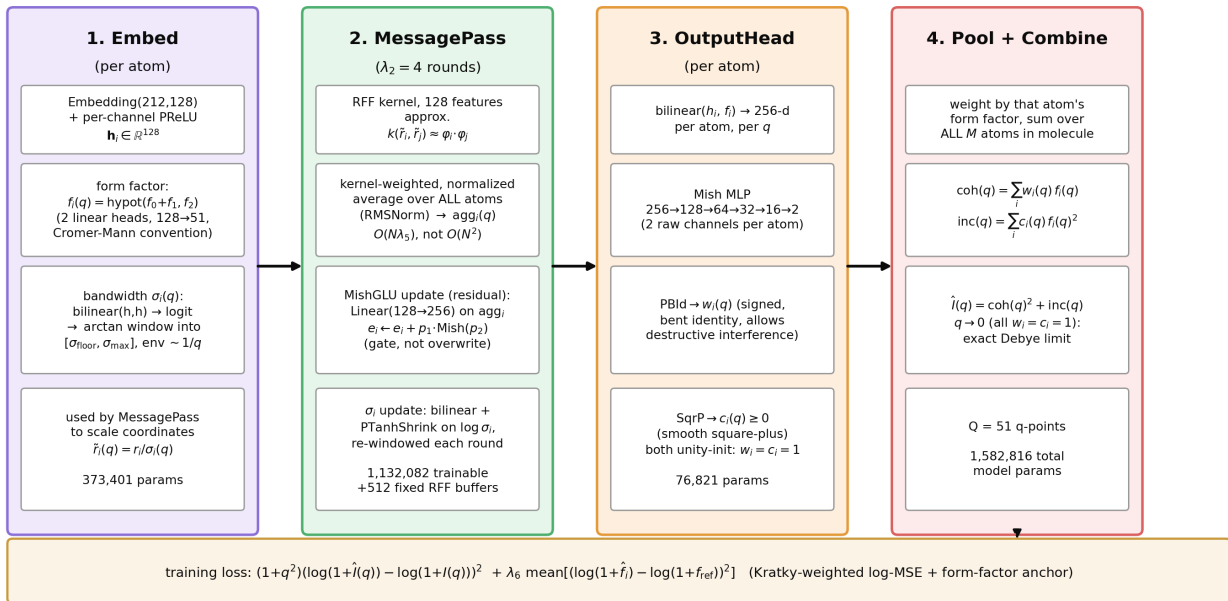


Figure 1: Input (atomic coordinates + element types) $\rightarrow$ ScatterNet $\rightarrow$ output $\hat{I}(q) = \text{coh}(q)^2 + \text{inc}(q)$: per-atom coherent (w_i , signed, captures destructive interference) and incoherent ($c_i \geq 0$) channels, weighted by each atom’s form factor and pooled over all atoms in the molecule.

3 Background & Related Work

CRY SOL [1] established the analytic and non-learned spherical-harmonic decomposition this project uses for ground truth, built on the Stuhmann expansion [2] of the exact Debye sum [3]. Graph neural networks for molecular property prediction were pioneered by SchNet [4]. DimeNet [5] added bond-angle information, improving accuracy on the QM9 dataset [6], part of which is folded into the training set of ScatterNet. PaiNN [7] extended message passing to rotationally equivariant properties. None of these five target $I(q)$ directly as a learned, differentiable quantity; ScatterNet replaces the $O(N^2)$ analytic sum with an $O(N)$ learned surrogate.

4 Data Processing

A dataset of 1,044,583 molecules spanning 1–6,046 atoms was assembled from 13 sources (Table 1), covering 474 unique atom/ion types. Raw structures (.cif, .pdb, .xyz) were converted to a uniform

Cartesian XYZ representation; dummy sites were removed, deuterium mapped to hydrogen, zero-atom structures discarded, and non-element placeholders (X, M, Cn) excluded from the ionic makeup. $I(q)$ for each molecule were calculated, and an ‘encoding’ was calculated, corresponding to the atom’s ionic makeup.

Table 1: Cleaned dataset composition by source.

Group	Molecules	Atom range
COD (Crystallography Open Database)	532,302	1–6,032
QM9 (small organic molecules)	133,844	3–29
tmQM (transition metal complexes)	108,541	7–569
rcsb_sml (PDB, small)	96,158	28–6,036
viro3D (viral protein structures)	60,488	173–6,046
hydration_shells (water solvation)	48,571	3–147
rcsb_med (PDB, medium)	31,749	2,996–6,046
mofs (metal-organic frameworks)	30,863	10–5,760
monoatomic/binary/Si-Ge/Ar-Ne/NaCl clusters	2,067	2–1,482
Total	1,044,583	1–6,046

$I(q)$ ground truth was computed from the cleaned XYZ via the Stuhmann decomposition at $L=50$ and stored, together with the coordinates, in a compressed HDF5 database. To avoid padding waste across a 1–6,046 atom size range, molecules are grouped by size into 57 atom-count buckets; each bucket is stratified 70/15/15 at the molecule level into train/val/test

independently, so every split contains both small and large molecules.

5 Architecture

ScatterNet (Fig. 1) has three stages, $Q=51$ q-points and a 212-entry element vocabulary (211 ions + 1 padding index). **Embed** (373,401 params) looks up a learned $\lambda_1=128$ -d vector per atom (Embedding(212, 128)) through a channel-wise **PReLU**, then splits it via two linear heads into a complex form-factor magnitude $f_i(q) = \text{hypot}(f_0+f_1, f_2)$ and, via a (128, 51, 51) bilinear layer, a kernel-bandwidth logit squashed into $(\sigma_{\text{floor}}, \sigma_{\text{max}})$ by a smooth **arctan** soft window, which was chosen over a hard clamp so gradients do not vanish at the boundary. **MessagePass** ($\lambda_2=4$ rounds, 1,132,082 trainable params + 512 fixed buffers) approximates the Radial Basis Function (RBF) $k(\tilde{r}_i, \tilde{r}_j) \approx \varphi_i \cdot \varphi_j$ between every atom pair via $\lambda_5=128$ fixed Random Fourier Features, cutting the all-pairs cost from $O(N^2)$ to $O(N\lambda_5)$. Each round, an atom’s embedding is updated by a kernel-weighted, RMSNorm-normalized average over all atoms, fed through a **Mish**-gated GLU (128→256, avoids dead units under early, noisy gradients) and added as a residual, $e_i \leftarrow e_i + \text{gate}_i$. In parallel, a bilinear layer updates $\log \sigma_i$ each round via **PTanhShrink** ($f(y)=y - c \tanh(y/c)$, learned width c , shrinks small noisy updates toward zero), re-windowed by the same arctan squash as Embed. **OutputHead** (76,821 params) bilinearly combines each atom’s updated embedding with its own $f_i(q)$, narrows through a Mish MLP (256→128→64→32→16→2) to two channels, then **PBId** gives $w_i(q)$ and **SqrP** (square-plus, strictly ≥ 0) gives $c_i(q)$; both unity-initialized ($w_i=c_i=1$). Only in the final **pooling** step are atoms combined: each channel is weighted by that atom’s own form factor and summed over all M atoms, $\text{coh}(q) = \sum_i w_i(q) f_i(q)$, $\text{inc}(q) = \sum_i c_i(q) f_i(q)^2$, giving $\hat{I}(q) = \text{coh}(q)^2 + \text{inc}(q)$. **Total: 1,582,816 parameters** (1,582,304 trainable + 512 fixed).

Training minimizes a Kratky-weighted log-MSE, $(1+q^2)(\log(1+\hat{I}(q)) - \log(1+I(q)))^2$, plus a form-factor penalty ($\lambda_6=1.0$, anchoring $\hat{f}_i(q)$ to tabulated xraydb values), via AdamW (lr = 1.3×10^{-4} , weight decay 0.1, ReduceLROnPlateau, grad clip 1.0), tensor/data-parallel routed across $2 \times \text{T4 GPUs}$ by bucket atom-count (dp_atom_threshold=101).

6 Baseline Model

Nine non-GNN baselines were implemented: closed-form physics fits per-molecule (Guinier, Guinier-Porod, a binned Debye sum over the histogrammed pairwise-distance distribution $P(r)$, recovering the exact Debye equation as bin count grows), stochastic estimators subsampling the $O(N^2)$ pairwise sum (propoEst, stratEst), and learned baselines with no 3-D access (a training-mean-by-atom-count lookup, a linear SVM over composition, a 2-layer MLP over composition + size). Beating Binned Debye (closest coarse-graining of the ground truth, thus a feasibility ceiling) is not the bar;

beating the composition-only baselines is, since that proves genuine 3-D structure use rather than size or composition alone.

7 Quantitative Results

Training reached epoch 33 (Fig. 2); test-set MSLE = 0.115 at 71.1 μ s/atom inference (Table 2). This already beats every composition-only baseline (Atom-Count, Linear SVM, MLP, Nearest-Neighbour) and the Guinier physics fits, confirming the model learning genuine 3-D structure. It still trails Binned Debye and stratEst, the two baselines with explicit pairwise-distance access. Only 5% of the dataset is drawn per epoch. An epoch trained on 100% of the dataset took around 12 hours to complete on the available 2 $\times$ T4 GPUs. This may explain the visible plateaus and short reversals in Fig. 2 (e.g. epochs 10–11, 14, 18).

Table 2: ScatterNet (epoch 33) vs. baseline anchors, by MSLE.

Model	MSLE
Binned Debye (ceiling)	0.0053
SAXS stratEst	0.0531
ScatterNet (epoch 33)	0.115
Guinier-Porod	0.346
SAXS propoEst	0.366
Linear SVM	0.772
Atom-Count	1.010
MLP	1.821

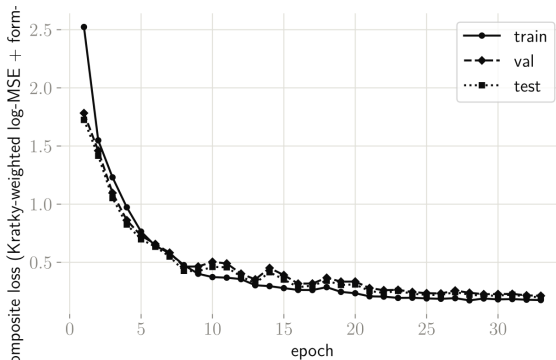
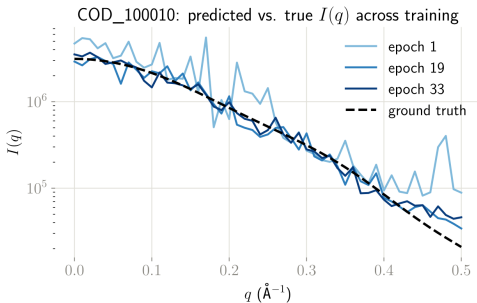
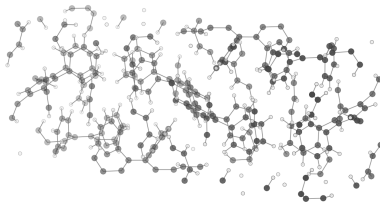


Figure 2: Composite train/val/test loss across 33 epochs.



(a) predicted vs. true $I(q)$



(b) COD_100010 (448 atoms)

Figure 3: (a) Predicted vs. true $I(q)$ for the crystal structure in (b), epochs 1/19/33; whole-curve mean log-space error falls monotonically, 0.55 $\rightarrow$ 0.20 $\rightarrow$ 0.19 (relative error 117% $\rightarrow$ 21.5% $\rightarrow$ 21.3%), tracking ground truth closely by epoch 19; only $q=0$ itself is non-monotonic (50% $\rightarrow$ 7% $\rightarrow$ 13%), consistent with Fig. 2’s per-epoch sampling noise. (b) A unit cell ($C_{264}H_{184}$), several distinct aromatic organic molecules.

8 Qualitative Results

Fig. 3(b) shows predicted-vs-true $I(q)$ convergence over training on a held-out crystal structure. Per- q error across the full test set follows the same pattern (Fig. 4).

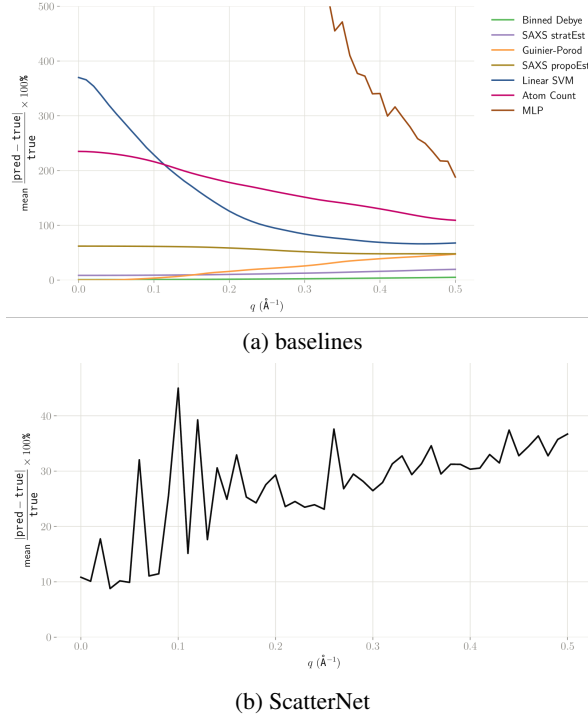


Figure 4: Per- q mean % error: baselines (a) vs. ScatterNet (b, epoch 33). Mean per- q $R^2=0.965$ (worst 0.917); mean error 27.0%, rising to 32.9% at high q .

9 Evaluate Model on New Data

See: Fig. 3’s structure ran through inference code independent of the training/val/test split.

10 Discussion

ScatterNet already beats every baseline that don’t compute pairwise distances, despite the baselines using 10% of the dataset whereas ScatterNet trained on 5%. It still trails Binned Debye and stratEst (Table 2); the gap concentrates in the low- q /Guinier-region magnitude (Fig. 3) and high- q fine structure (Fig. 4), not uniformly across the curve.

(channel drift): the coherent PBId bend

weight relaxed $0.98 \rightarrow 0.74$ (more linear, less bent); the incoherent SqrP width shrank $4.0 \rightarrow 2.23$ (sharper positivity floor); and the largest relative movement was in MessagePass’s later-round sigma-bilinear layers.

11 Ethical Considerations

All training data is open-source and fully attributed; no private/sensitive information is involved.

12 Project Difficulty / Quality

Three factors made this very difficult. First, The $O(N)$ ground-truth pipeline (Stuhrmann, not naive Debye) and a two-channel head whose $q \rightarrow 0$ limit is derived from the Debye equation (Fig. 1) required deriving the underlying physics, from first principles. Additionally, the math behind the RFF approximation and the resulting einsums which were needed to accumulate the sums were difficult to learn. Second, on $2 \times T4$ ’s 16GB/GPU ceiling, real runs OOM’d and ran extremely slow repeatedly from activation memory before any model issue were made. Fixing it took engineering chunking across multiple axes, tensor/data parallelism, implementing torch compile, and implementing a custom bilinear layer which avoids CUDA kernel launches. Third, even at 5% coverage, epochs take $\sim 1\text{hr}$ so every hyperparameter changes had to be deliberate and the majority of runs had to be terminated before completion. (Table 2).

References

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